

Customer Churn Analysis

1. Project Overview

This project focuses on analyzing customer churn patterns using a telecom customer dataset.

The goal was to understand the factors influencing churn, build a predictive machine-learning model, and visualize insights through an interactive **Power BI dashboard**.

The workflow involved **data cleaning in pandas**, **EDA**, **model training**, and finally storing the trained model in a **.pkl file** for future use.

2. Dataset Summary

- **Rows:** 7,043
 - **Columns:** Demographics, services subscribed, billing details, and churn status
 - **Key Features Include:**
 - **Customer Demographics:** Gender, SeniorCitizen, Partner, Dependents, Tenure
 - **Service Details:** InternetService, OnlineSecurity, OnlineBackup, TechSupport, StreamingTV, PhoneService
 - **Billing Details:** MonthlyCharges, TotalCharges, PaperlessBilling, Contract
 - **Target Variable:** Churn (Yes/No)
 - **Data Issues Identified:**
 - Missing TotalCharges values
 - Incorrect data types (TotalCharges stored as string)
 - Mixed categorical encodings
-

3. Exploratory Data Analysis using Python (pandas)

Python and pandas were used for data cleaning, transformation, and exploratory analysis.

Data Preparation

- Loaded dataset using pandas
- Used df.info() and df.describe() to examine structure
- Converted **TotalCharges** to numeric
- Removed or imputed missing values
- Standardized column names where needed

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", "0.0")
```

```
df["TotalCharges"] = df["TotalCharges"].astype(float)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 20 columns):
#   Column                Non-Null Count  Dtype
---  -
0   gender                 7043 non-null   object
1   SeniorCitizen          7043 non-null   int64
2   Partner                7043 non-null   object
3   Dependents             7043 non-null   object
4   tenure                 7043 non-null   int64
5   PhoneService           7043 non-null   object
6   MultipleLines           7043 non-null   object
7   InternetService        7043 non-null   object
8   OnlineSecurity         7043 non-null   object
9   OnlineBackup           7043 non-null   object
10  DeviceProtection       7043 non-null   object
11  TechSupport            7043 non-null   object
12  StreamingTV            7043 non-null   object
13  StreamingMovies        7043 non-null   object
14  Contract               7043 non-null   object
15  PaperlessBilling       7043 non-null   object
16  PaymentMethod          7043 non-null   object
17  MonthlyCharges         7043 non-null   float64
18  TotalCharges           7043 non-null   float64
19  Churn                  7043 non-null   object
dtypes: float64(2), int64(2), object(16)
memory usage: 1.1+ MB
```

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

Feature Engineering

- Created **tenure groups** for better customer segmentation
- Label-encoded categorical variables via **encoders.pkl**
- Split data into feature matrix and target labels

Key Insights from EDA

- **Month-to-month contract** customers show the highest churn
- **Electronic check** payment method has the highest churn rate
- **Low tenure customers (0–3 months)** churn significantly
- Customers with **higher MonthlyCharges** exhibit higher churn
- Gender has minimal impact on churn (as seen in your dashboard image)

These insights align with the visual patterns displayed in your Power BI dashboard image.

4. Machine Learning Model Training

A machine learning model was built to predict whether a customer will churn.

Model Details

- **Algorithm Used:** RandomForestClassifier
- **Train–Test Split:** Standard 70–30 split
- **Preprocessing:** Label encoding + numeric conversions
- **Model Evaluation:** Accuracy, precision, recall, and churn recall were prioritized
- **Deployment Files Generated:**
 - customer_churn_model.pkl – stores the trained model
 - encoders.pkl – stores feature encoders for reproducible prediction

The saved PKL files allow seamless integration with future applications such as a churn prediction API or a web dashboard.

```
rfc.fit(x_train_smote, y_train_smote)
```

▼ RandomForestClassifier ⓘ ?

► Parameters

```
#evaluate on test data
y_test_pred = model.predict(x_test)
print("Accuracy Score:\n", accuracy_score(y_test, y_test_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_test_pred))
print("Classification Report:\n", classification_report(y_test, y_test_pred))
```

Accuracy Score:

0.7771469127040455

Confusion Matrix:

[[880 156]

[158 215]]

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.85	0.85	1036
1	0.58	0.58	0.58	373
accuracy			0.78	1409
macro avg	0.71	0.71	0.71	1409
weighted avg	0.78	0.78	0.78	1409

5. Power BI Interactive Dashboard

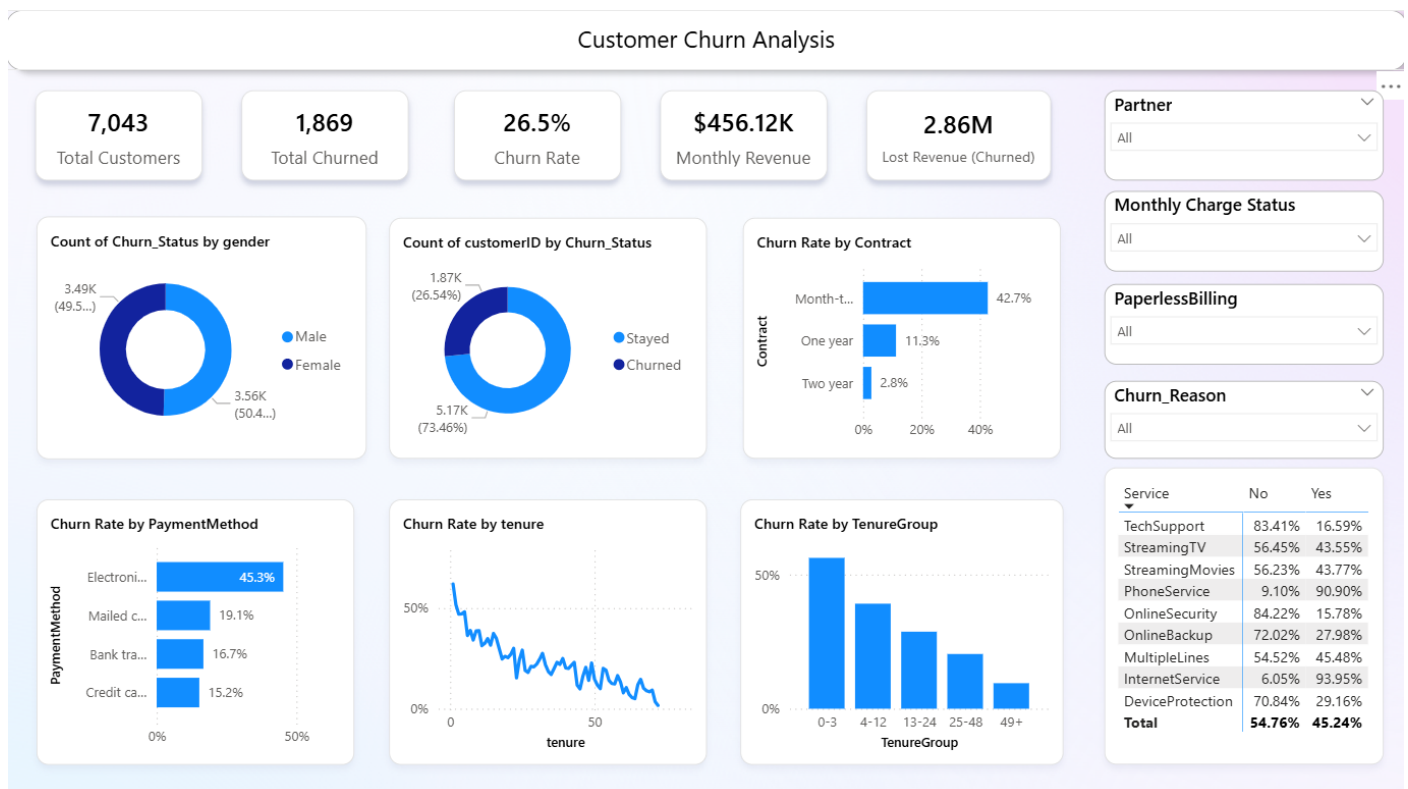
An interactive churn analysis dashboard was created in Power BI using the cleaned dataset. The dashboard includes KPI cards, donut charts, bar charts, and trendlines.

Dashboard Highlights (based on your uploaded image):

- **KPIs:**
 - 7,043 Total Customers
 - 1,869 Total Churned
 - 26.5% Churn Rate
 - \$456.12K Monthly Revenue
 - 2.86M Lost Revenue
- **Visuals Include:**
 - Churn by Gender (Donut Chart)
 - Churned vs Stayed Customers

- Churn Rate by Contract Type
- Churn Rate by Payment Method
- Tenure vs Churn Trendline
- Churn Rate by Tenure Group
- Service-wise churn comparison table

The dashboard supports filtering by **Partner**, **Paperless Billing**, **Churn Reason**, etc., enabling deep exploration of churn behavior.



6. Business Recommendations

Based on the results from EDA, model predictions, and dashboard insights:

1. Promote Long-Term Contracts

Long-term (1–2 year) contract customers show significantly lower churn. Offer incentives to migrate month-to-month customers to annual plans.

2. Improve Payment Experience

Electronic Check users have the highest churn. Introduce smoother digital payment alternatives and loyalty perks.

3. Target Early-Tenure Customers

Customers with <6 months tenure churn at the highest rate. Initiate early onboarding, welcome benefits, and personalized support.

4. Strengthen Add-on Services

Customers lacking OnlineSecurity, TechSupport, or Backup see higher churn. Feature these add-ons in retention offers.

5. Retain High-Revenue Customers

Focus retention strategies on customers with high MonthlyCharges and long service usage.